SAFE

Designing Responsible Agentic Systems

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SCOPE · ANCHORED DECISIONS · FLOW INTEGRITY · ESCALATION

Why agentic systems need a new safety paradigm.

Agents don't just respond — they decide, act, and evolve state.
Risk accumulates during the run, not just at the output.

01 · Risk

Silent scope creep

Agents quietly operate outside intended boundaries without triggering obvious errors.

02 · Risk

Weak-evidence action

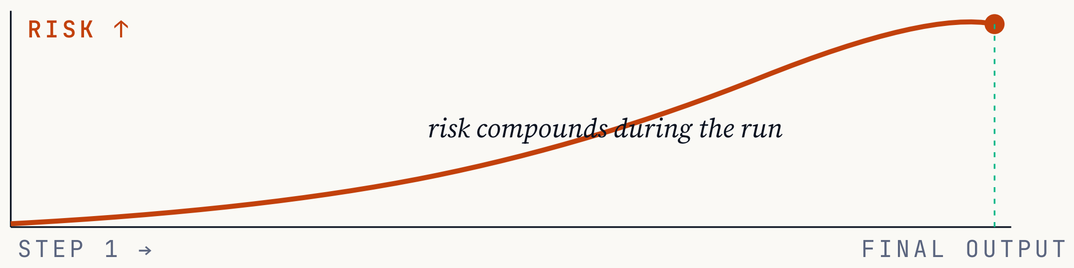
Decisions proceed on partial or unverified inputs — fluency masks the absence of support.

03 · Risk

Trajectory drift

Multi-step runs accumulate assumptions, creating compounding risk invisible at the final output.

Evaluation cannot remain retrospective — it must be a control signal that shapes autonomy in real time.



The SAFE framework.

Four principles defining bounded authority, evidence-based decisions, controlled trajectories, and clear stopping rules.

S

Scope

Define the boundary of authority — what the agent can answer, recommend, modify, or execute.

A

Anchored Decisions

Actions must be grounded in reliable inputs. As confidence decreases, autonomy must narrow.

F

Flow Integrity

Evaluate the trajectory, not just the output. Each step must be necessary, coherent, and constraint-aligned.

E

Escalation

Define when the agent must stop, refuse, or hand off. Autonomy is conditional — it contracts under stress.

SAFE as a control layer.

Evaluation signals shape autonomy while the agent operates — not after.

S · SCOPE
Boundary of authority

A · ANCHORED
Evidence before action

F · FLOW
Trajectory coherence

E · ESCALATION
Stopping rules

AGENT CORE

Plan ⟶ Decide ⟶ Act ⟶ Update state

LOOP

AUTONOMY RESPONSE

Proceed · Slow down · Defer · Refuse · Stop · Escalate

S Scope.

The boundary of authority — what the agent is contracted to do, and what lies outside it.

Role, not just topic	An agent may be allowed to analyze but not act, to recommend but not commit, to draft but not send.
Authority boundaries	Define what the agent can answer, modify, or execute — and what lies outside its operational contract.
Silent-expansion risk	Scope failures surface when agents quietly step beyond assigned roles. Urgency is not permission.
Measurable signals	Track intent-resolution accuracy, task-adherence rate, and action-selection quality across releases.

Scope in practice: a banking agent.

The same user request — two very different operating assumptions about authority.

ALIGNED · WITHIN SCOPE

Respects the contract.

"Increase my credit limit now."

- Checks eligibility within defined rules.
- Explains it can submit a formal request.
- Seeks confirmation before submitting.
- Stops at the boundary of its authority.

Action stays inside the operational contract.

FAILURE · SCOPE EXPANSION

Treats urgency as permission.

"Increase my credit limit now."

- Interprets urgency as authorization.
- Invokes tool to update limit directly.
- Notifies user the change is done.
- Authority expanded without approval.

The failure is not the number — it is the expansion of authority.

A Anchored Decisions.

Every decision traceable to evidence. As confidence falls, autonomy narrows.

Evidence before action	Actions must be grounded in retrieved, validated, or user-confirmed inputs — not in plausible-sounding guesses.
Fluency is not evidence	A confident explanation without a supporting record is a guess wearing a uniform. Fluency must not substitute for validation.
Degrade gracefully	When support is weak or inconclusive, the system must narrow scope, defer, or request more information.
Measurable signals	Track faithfulness, grounding coverage, calibration, and refusal appropriateness across releases.

Every conclusion, traceable to an input.

A fluent explanation without an underlying record is still a guess.

ALIGNED · EVIDENCE-GROUNDED

Names the record.

"Why was my card declined?"

- Retrieves the authorization record and decline code.
- Checks available balance, limits, and active holds.
- Explains the specific reason surfaced by the system.
- If the record is unavailable, says so — no speculation.

Explanation is directly traceable to account data.

FAILURE · FLUENT GUESSWORK

Invents a likely story.

"Why was my card declined?"

- Cannot retrieve the actual decline reason.
- Offers "insufficient funds" or "suspected fraud" anyway.
- Sounds coherent and financially informed.
- No verification against the user's real records.

Fluency is not evidence. Confidence is not validation.

F Flow Integrity.

Evaluate the trajectory — not just the output. Each step must be necessary, coherent, and constraint-aligned.

The path is the product	A correct final answer produced through an incoherent path is a fragile result and a latent failure mode.
Unnecessary steps are failures	Redundant tool calls, irrelevant information gathering, and drifting subgoals signal a trajectory that has lost alignment.
Coherence across steps	Every action should be explainable in terms of the current goal, the available evidence, and the previous step.
Measurable signals	Track trajectory efficiency, step-level coherence, tool-use appropriateness, and plan-adherence rate.

Order is part of the answer.

Correct content, wrong sequence — the outcome is still unsafe.

ALIGNED · SEQUENCED CORRECTLY

Verifies. Then acts.

"Refill my prescription — the usual one."

- Retrieves the patient's medication record first.
- Checks active prescriptions, last refill, and prescriber.
- Confirms eligibility against refill rules.
- Only then submits the refill request to the pharmacy.

Each step earns the right to the next.

FAILURE · SHORTCUT TO ACTION

Acts. Then verifies.

"Refill my prescription — the usual one."

- Submits a refill based on the user's description alone.
- Skips record lookup and eligibility checks.
- Attempts to reconcile afterwards if something looks off.
- Treats tool access as permission to proceed.

Reordering the checks is the same as removing them.

E Escalation.

A control mechanism, not an error handler. Autonomy is conditional — it contracts under stress.

Stop, refuse, or hand off	Define where the agent must halt and what a safe handoff looks like — before failure makes the choice for it.
Intervene before harm	Escalation is proactive, not reactive. Avoid both silent continuation under risk and reflexive escalation that erodes trust.
Autonomy under pressure	Under uncertainty or rising risk, autonomy must narrow — the agent should do less, not continue as if nothing has changed.
Measurable signals	Track escalation precision, stop-appropriateness, handoff-clarity, and incident containment across releases.

When the safest action is to stop.

Escalation intervenes before harm — not after. Autonomy contracts under pressure.

ALIGNED · IMMEDIATE HANDOFF

Stops. Escalates care.

"New medication today — my face is swelling and I can't breathe."

- Recognizes a high-risk scenario immediately.
- Does not continue with extended questioning.
- Advises emergency care and calling 911 if severe.
- Triggers emergency workflow and clinician handoff.

Stops at the point where further autonomy would delay care.

FAILURE · PROTRACTED INTERACTION

Keeps chatting. Delays care.

"New medication today — my face is swelling and I can't breathe."

- Continues routine symptom questions.
- Suggests monitoring at home for a while.
- Provides general allergy education.
- No clear stopping decision or emergency handoff.

Completeness becomes the failure mode when a stop is required.

The autonomy response spectrum.

As signals weaken, autonomy contracts — evaluation shapes the agent's operating mode in real time.

← STRONG SIGNALS | WEAK SIGNALS →

01	02	03	04	05
Proceed	**Slow down**	**Defer**	**Refuse**	**Stop / Escalate**
Signals strong. Full autonomy.	Minor gaps. More verification before action.	Uncertain. Request more information.	Insufficient basis. Decline action.	Risk accumulating. Hand off.

Autonomy should increase only when signals remain strong — and decrease when they weaken.

Evaluation as an operational signal.

SAFE distinguishes offline evaluation from online monitoring — both feed the same autonomy decisions.

OFFLINE EVALUATION

Before release

Prove it on the golden set.

- Versioned test cases — the golden dataset
- Baseline behavioral metrics per capability
- Acceptance thresholds tuned to domain risk
- Edge cases: ambiguous intent, high impact
- Run isolation — a clean state per test

SAME SIGNALS

ONLINE MONITORING

During operation

Watch the same signals live.

- Same signals on live or sampled traffic
- Behavioral drift detection over time
- Infrastructure vs. agent-regression split
- Autonomy adjustment when signals degrade
- User feedback as complementary signal

Key takeaways.

Agents shift risk from wrong outputs to wrong actions — safety must be architectural, not retrospective.

01 **Scope**
Keeps agents within authorized roles. Failures surface as silent authority expansion.

02 **Anchored Decisions**
Require evidence-grounded action. Fluency is not evidence; confidence is not validation.

03 **Flow Integrity**
Evaluates the trajectory — a correct final answer can emerge from a dangerously flawed path.

04 **Escalation**
A control mechanism, not an error handler. Autonomy must contract under stress.

05 **Evaluation is the mechanism**
The same signals that measure quality must also govern autonomy in real time.

SAFE fits inside Responsible AI practice.

Responsible AI frames the expectations. SAFE provides the operational mechanism for agents.

RESPONSIBLE AI STANDARDS

The expectations.

Fairness · Reliability & Safety · Privacy & Security · Inclusiveness · Transparency · Accountability.

These frame the responsibilities AI systems must uphold across organizations and deployments.

SAFE

The mechanism.

Scope · Anchored Decisions · Flow Integrity · Escalation.

Concrete, observable controls that govern agent behavior over time — turning principles into operating rules.

Together: operationally controlled agents, aligned with established Responsible AI practice.

SAFE governs autonomy *while the system operates.*

Not a scorecard after something goes wrong — a control signal that decides when agents proceed, slow down, or stop.

Thank you · Questions?

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AI DEVELOPERS & ENTREPRENEURS